

CHEMISTRY

46. If the molar conductivity (Λ_m) of a 0.050 mol L^{-1} solution of a monobasic weak acid is $90 \text{ S cm}^2 \text{ mol}^{-1}$, its extent (degree) of dissociation will be

[Assume $\Lambda_+^\circ = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$ and $\Lambda_-^\circ = 50.4 \text{ S cm}^2 \text{ mol}^{-1}$.]

- (1) 0.215 (2) 0.115
(3) 0.125 (4) 0.225

Answer (4)

Sol. Degree of dissociation (α) is given as

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

$$\Lambda_m^\circ = \Lambda_+^\circ + \Lambda_-^\circ$$

$$= 349.6 + 50.4$$

$$= 400 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{90}{400} = 0.225$$

47. Given below are two statements :

Statement I : A hypothetical diatomic molecule with bond order zero is quite stable.

Statement II : As bond order increases, the bond length increases.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Both Statement I and Statement II are false
(4) Statement I is true but Statement II is false

Answer (3)

Sol.

- A positive bond order means a stable molecule while a negative or zero bond order means an unstable molecule.
- When bond order increases, the bond length decreases.

48. The ratio of the wavelengths of the light absorbed by a Hydrogen atom when it undergoes $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$ transitions, respectively, is

- (1) $\frac{1}{4}$ (2) $\frac{1}{36}$
(3) $\frac{1}{16}$ (4) $\frac{1}{9}$

Answer (1)

Sol. $\Delta E = \frac{hc}{\lambda} = E_{\text{final}} - E_{\text{initial}} \quad \left(E_n = \frac{-R_H}{n^2} \right)$

$$\begin{aligned}\Delta E_{2 \rightarrow 3} &= \frac{hc}{\lambda_{2 \rightarrow 3}} = E_3 - E_2 = \frac{-R_H}{3^2} - \left(\frac{-R_H}{2^2} \right) \\ &= R_H \left(\frac{1}{4} - \frac{1}{9} \right) \\ &= R_H \times \frac{5}{36}\end{aligned}$$

$$\therefore \lambda_{2 \rightarrow 3} = \frac{hc \cdot 36}{R_H \cdot 5}$$

$$\Delta E_{4 \rightarrow 6} = E_6 - E_4 = \frac{-R_H}{36} + \frac{R_H}{16} = \frac{R_H \times 20}{36 \times 16}$$

$$\frac{hc}{\lambda_{4 \rightarrow 6}} = \frac{R_H \times 20}{36 \times 16}$$

$$\lambda_{4 \rightarrow 6} = \frac{hc \times 36 \times 16}{R_H \cdot 20}$$

$$\frac{\lambda_{2 \rightarrow 3}}{\lambda_{4 \rightarrow 6}} = \frac{\frac{hc \cdot 36}{R_H \cdot 5}}{\frac{hc \times 36 \times 16}{R_H \cdot 20}}$$

$$= \frac{1}{4}$$

49. The correct order of the wavelength of light absorbed by the following complexes is,

- A. $[\text{Co}(\text{NH}_3)_6]^{3+}$ B. $[\text{Co}(\text{CN})_6]^{3-}$
C. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ D. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$

Choose the **correct** answer from the options given below:

- (1) $C < A < D < B$
(2) $B < D < A < C$
(3) $B < A < D < C$
(4) $C < D < A < B$

Answer (3)

Sol. $\lambda \propto \frac{1}{\text{strength of ligand}}$

$$\lambda \propto \frac{1}{\text{splitting}}$$

- A. $[\text{Co}(\text{NH}_3)_6]^{3+}$; 475 nm
B. $[\text{Co}(\text{CN})_6]^{3-}$; 310 nm
C. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$; 600 nm
D. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$; 498 nm

Order of $\lambda = C > D > A > B$

50. If the rate constant of a reaction is 0.03 s^{-1} , how much time does it take for 7.2 mol L^{-1} concentration of the reactant to get reduced to 0.9 mol L^{-1} ?

(Given: $\log 2 = 0.301$)

- (1) 21.0 s (2) 69.3 s
(3) 23.1 s (4) 210 s

Answer (2)

Sol. $k = 0.03 \text{ s}^{-1}$

$$\begin{aligned}
 t &= \frac{2.303}{k} \log \frac{a}{a-x} \\
 &= \frac{2.303}{0.03} \log \frac{7.2}{0.9} \\
 &= \frac{2.303}{0.03} \log 8 \\
 &= \frac{2.303}{0.03} \times 3 \times \log 2 \\
 &= \frac{2.303}{0.03} \times 3 \times 0.301 \\
 &= 69.3 \text{ s}
 \end{aligned}$$

51. Match **List I** with **List II**

	List-I (Mixture)		List-II (Method of separation)
A.	$\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	(I)	Distillation under reduced pressure
B.	Crude oil in petroleum industry	(II)	Steam distillation
C.	Glycerol from spent-lye	(III)	Fractional distillation
D.	Aniline - water	(IV)	Simple distillation

Choose the **correct** answer from the options given below:

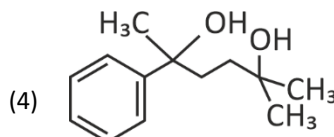
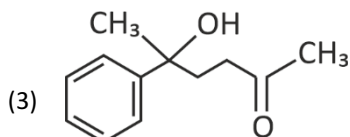
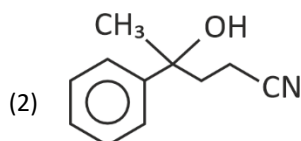
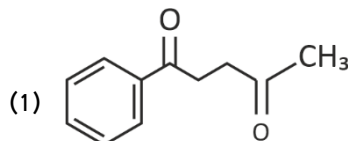
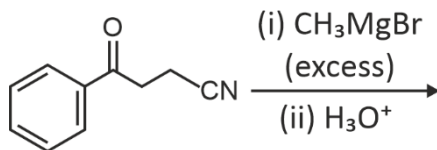
- (1) A-III, B-IV, C-II, D-I (2) A-IV, B-III, C-I, D-II
(3) A-IV, B-III, C-II, D-I (4) A-III, B-IV, C-I, D-II

Answer (2)

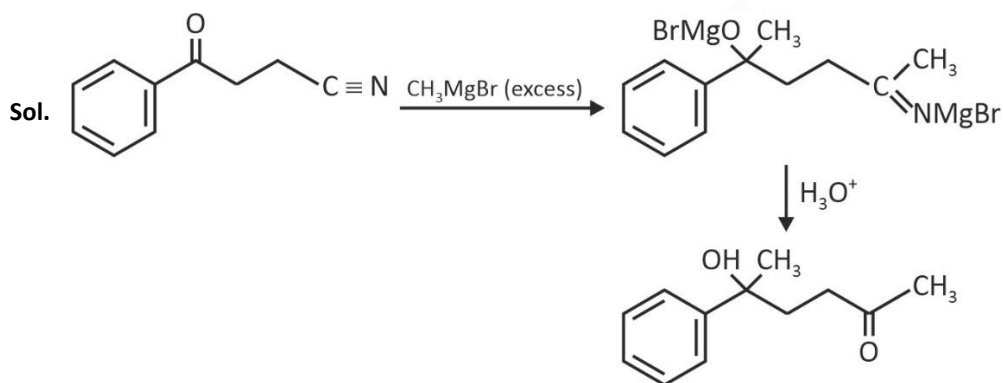
Sol.

		(Method of separation)
(A)	$\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	Simple distillation
(B)	Crude oil in petroleum industry	Fractional distillation
(C)	Glycerol from spent-lye	Distillation under reduced pressure
(D)	Aniline - water	Steam Distillation

52. The major product of the following reaction is



Answer (3)



53. Which one of the following compounds can exist as cis-trans isomers?

(1) 1, 2-Dimethylcyclohexane

(2) Pent-1-ene

(3) 2-Methylhex-2-ene

(4) 1, 1-Dimethylcyclopropane

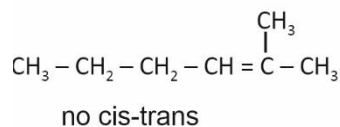
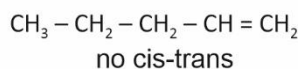
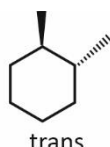
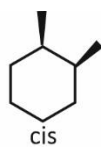
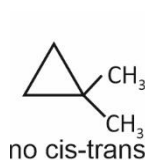
Answer (1)

Sol. Cis-trans isomers shown by :

Condition: Restricted rotation around double bond

Or

Different group around double bond



54. Among the following, choose the ones with equal number of atoms.

- A. 212 g of $\text{Na}_2\text{CO}_3(\text{s})$ [molar mass = 106 g]
 B. 248 g of $\text{Na}_2\text{O}(\text{s})$ [molar mass = 62 g]
 C. 240 g of $\text{NaOH}(\text{s})$ [molar mass = 40 g]
 D. 12 g of $\text{H}_2(\text{g})$ [molar mass = 2 g]
 E. 220 g of $\text{CO}_2(\text{g})$ [molar mass = 44 g]

Choose the **correct** answer from the options given below :

- (1) B, D, and E only (2) A, B, and C only
 (3) A, B, and D only (4) B, C, and D only

Answer (3)

Sol. Number of atoms = $\frac{\text{given mass}}{\text{molar mass}} \times \text{atomicity} \times N_A$

A. $\frac{212}{106} \times 6 \times N_A = 12 N_A$

B. $\frac{248}{62} \times 3 \times N_A = 12 N_A$

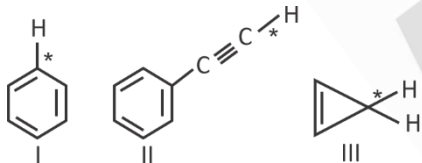
C. $\frac{240}{40} \times 3 \times N_A = 18 N_A$

D. $\frac{12}{2} \times N_A \times 2 = 12 N_A$

E. $\frac{220}{44} \times N_A \times 3 = 15 N_A$

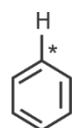
A, B and D have same number of atoms

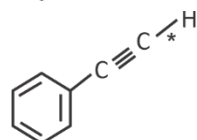
55. Among the given compounds I-III, the correct order of bond dissociation energy of C-H bond marked with * is :



- (1) II > III > I (2) II > I > III
 (3) I > II > III (4) III > II > I

Answer (2)

Sol. I.  carbon of this bond is sp^2 hybridised

II.  carbon of this bond is sp hybridised

III.  carbon of this bond is sp^3 hybridised

Higher the percentage s character, stronger is C-H bond. Correct order of bond dissociation energy of C-H bond:

II > I > III

56. The standard heat of formation, in kcal/mol of Ba^{2+} is :

[Given : standard heat of formation of SO_4^{2-} ion (aq) = -216 kcal/mol, standard heat of crystallisation of $\text{BaSO}_4(\text{s}) = -4.5$ kcal/mol, standard heat of formation of $\text{BaSO}_4(\text{s}) = -349$ kcal/mol]

- (1) $+220.5$ (2) -128.5
(3) -133.0 (4) $+133.0$

Answer (2)



From equation (1),(2) and (3) we get equation (4). Applying equation (3) – (1) – (2)

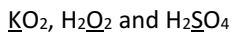
So, $-349 - (-4.5) - (-216)$

$\text{So } -349 + 4.5 + 216$

$= -349 + 220.5$

$= -128.5 \text{ kcal/mol}$

57. Consider the following compounds :



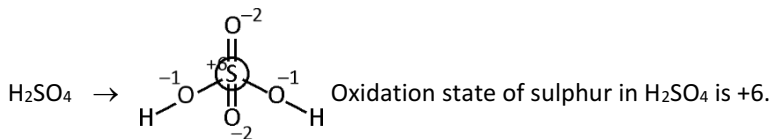
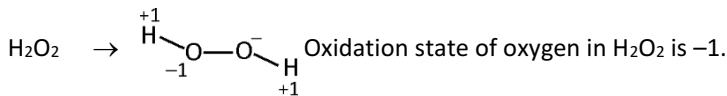
The oxidation state of the underlined elements in them are, respectively,

- (1) $+4, -4$, and $+6$ (2) $+1, -1$, and $+6$
(3) $+2, -2$, and $+6$ (4) $+1, -2$, and $+4$

Answer (2)

Sol.

$\text{KO}_2 \rightarrow$ Alkali metal always shows $+1$ oxidation state. Therefore oxidation state of K is $+1$.



58. Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?

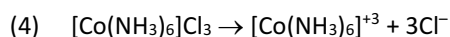
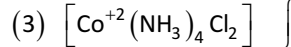
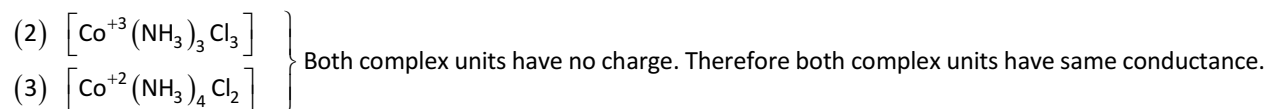
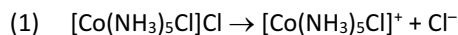
- (1) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}$ (2) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
(3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (4) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

Answer (2, 3)

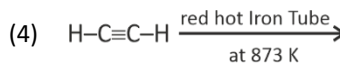
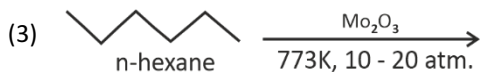
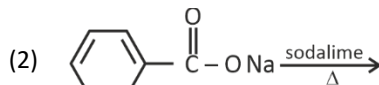
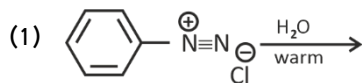
Sol. Conductance of any complex depends on the following factor.

(1) Number of ions produced by complex.

(2) If number of ions are same then we will check charge on complex unit.

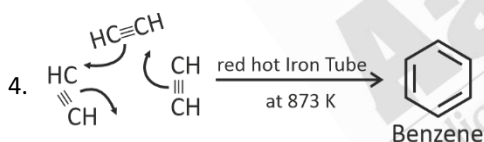
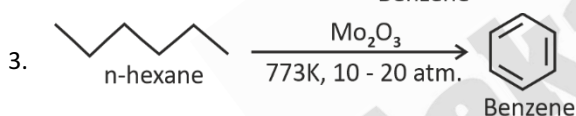
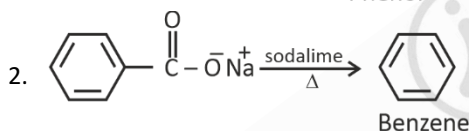
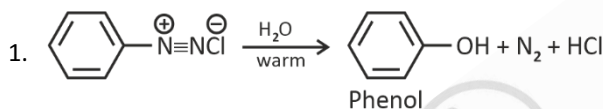


59. Which one of the following reactions does **NOT** give benzene as the product?

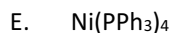
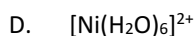
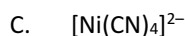


Answer (1)

Sol.



60. Which of the following are paramagnetic?



Choose the **correct** answer from the options given below :

(1) A, D and E only

(2) A and C only

(3) B and E only

(4) A and D only

Answer (4)

Sol.

A. $[\text{NiCl}_4]^{2-}$; Ni^{+2} ; $3d^8$; sp^3 hybridisation; 2 unpaired electrons; paramagnetic

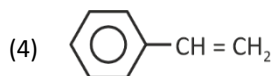
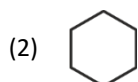
B. $\text{Ni}(\text{CO})_4$; Ni; $3d^8 4s^2$; sp^3 hybridisation; Zero unpaired electron; diamagnetic

C. $[\text{Ni}(\text{CN})_4]^{2-}$; Ni^{+2} ; $3d^8$; dsp^2 hybridisation; Zero unpaired electron; diamagnetic

D. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$; Ni^{2+} ; $3d^8$; sp^3d^2 hybridisation; Two unpaired electron; paramagnetic

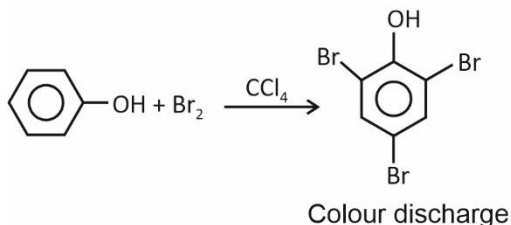
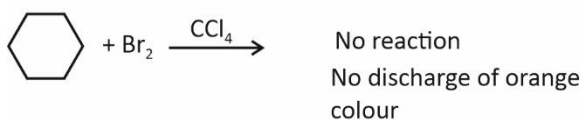
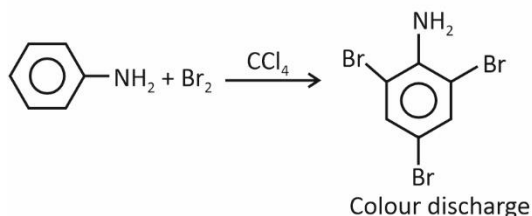
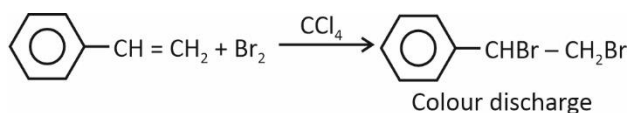
E. $\text{Ni}(\text{PPh}_3)_4$; Ni; $3d^8 4s^2$; sp^3 hybridisation; zero unpaired electron; Diamagnetic

61. Which one of the following compounds **does not** decolourize bromine water?



Answer (2)

Sol. Test for unsaturation i.e. Bromine water Reddish orange colour of bromine solution in CCl_4 will discharge when bromine adds to an unsaturation site.



62. Match **List-I** with **List-II**.

	List-I		List-II
A.	Haber process	I.	Fe catalyst
B.	Wacker oxidation	II.	PdCl_2
C.	Wilkinson catalyst	III.	$[(\text{PPh}_3)_3\text{RhCl}]$
D.	Ziegler catalyst	IV.	TiCl_4 with $\text{Al}(\text{CH}_3)_3$

Choose the **correct** answer from the options given below :

- (1) A-I, B-IV, C-III, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-III, C-I, D-IV
- (4) A-I, B-II, C-III, D-IV

Answer (4)**Sol.**

	Process		Catalyst used
A.	Haber process	I.	Fe catalyst
B.	Wacker oxidation	II.	PdCl_2
C.	Wilkinson catalyst	III.	$[(\text{PPh}_3)_3\text{RhCl}]$
D.	Ziegler catalyst	IV.	TiCl_4 with $\text{Al}(\text{CH}_3)_3$

63. Match **List-I** with **List-II**.

	List-I (Name of Vitamin)		List-II (Deficiency disease)
A.	Vitamin B_{12}	I.	Cheilosis
B.	Vitamin D	II.	Convulsions
C.	Vitamin B_2	III.	Rickets
D.	Vitamin B_6	IV.	Pernicious anaemia

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I (2) A-I, B-III, C-II, D-IV
 (3) A-IV, B-III, C-I, D-II (4) A-II, B-III, C-I, D-IV

Answer (3)**Sol.**

	List-I (Name of Vitamin)	List-II (Deficiency disease)
A.	Vitamin B_{12}	Pernicious anaemia
B.	Vitamin D	Rickets
C.	Vitamin B_2	Cheilosis
D.	Vitamin B_6	Convulsions

64. Given below are two statements :

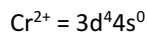
Statement I : Ferromagnetism is considered as an extreme form of paramagnetism.**Statement II** : The number of unpaired electrons in a Cr^{2+} ion ($Z = 24$) is the same as that of a Nd^{3+} ion ($Z = 60$).In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Statement I is false but Statement II is true (2) Both Statement I and Statement II are true
 (3) Both Statement I and Statement II are false (4) Statement I is true but Statement II is false

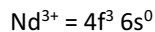
Answer (4)

Sol. Substances which are attracted very strongly in applied magnetic field are termed as ferromagnetic. Infact, ferromagnetism is an extreme form of paramagnetism.

Hence statement I is correct.



Unpaired electrons = 4



Unpaired electrons = 3

Hence, Statement II is incorrect

65. If the half-life ($t_{1/2}$) for a first order reaction is 1 minute, then the time required for 99.9% completion of the reaction is closest to :

- (1) 10 minutes (2) 2 minutes
(3) 4 minutes (4) 5 minutes

Answer (1)

Sol. For 1st order reaction

$$kt = 2.303 \log \frac{A_0}{A_t} \quad A_0 = \text{initial concentration}$$

A_t = Final concentration

$$t_{99.9\%} = 10 t_{1/2}$$

$$t_{99.9\%} = 10 \times 1 \text{ minute} = 10 \text{ minutes}$$

66. The correct order of decreasing basic strength of the given amines is:

- (1) benzenamine > ethanamine > N-methylaniline > N-ethylethanamine
(2) N-methylaniline > benzenamine > ethanamine > N-ethylethanamine
(3) N-ethylethanamine > ethanamine > benzenamine > N-methylaniline
(4) N-ethylethanamine > ethanamine > N-methylaniline > benzenamine

Answer (4)

Sol. Lower is the value of pK_b , higher is the basicity

Also aliphatic amines are stronger bases than aromatic amines.

pK_b : Benzenamine > N-Methylaniline > Ethanamine > N-Ethylethanamine

Basic strength : N-Ethylethanamine > Ethanamine > N-Methylaniline > Benzenamine

67. Match **List I** with **List II**

	List-I (Ion)		List-II (Group Number in Cation Analysis)
A.	Co^{2+}	I.	Group-I
B.	Mg^{2+}	II.	Group-III
C.	Pb^{2+}	III.	Group-IV
D.	Al^{3+}	IV.	Group-VI

Choose the **correct** answer from the options given below :

(1) A-III, B-II, C-I, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-II, C-IV, D-I

Answer (3)**Sol.**

	Ion	Group number in Cation Analysis
A.	Co^{2+}	Group-IV
B.	Mg^{2+}	Group-VI
C.	Pb^{2+}	Group-I
D.	Al^{3+}	Group-III

68. Phosphoric acid ionizes in three steps with their ionization constant values K_{a_1} , K_{a_2} and K_{a_3} , respectively, while K is the overall ionization constant. Which of the following statements are true?

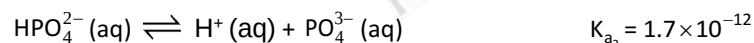
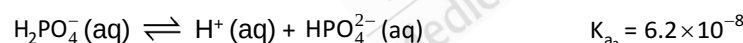
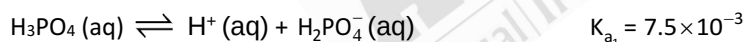
A. $\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$ B. H_3PO_4 is a stronger acid than H_2PO_4^- and HPO_4^{2-} C. $K_{a_1} > K_{a_2} > K_{a_3}$ D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$ Choose the **correct** answer from the options given below :

(1) A, B and C only

(2) A and B only

(3) A and C only

(4) B, C and D only

Answer (1)**Sol.** H_3PO_4 is a stronger acid than H_2PO_4^- and HPO_4^{2-} 

$$K_{a_1} > K_{a_2} > K_{a_3}$$

$$\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$$

Ans. (A), (B) and (C) only

69. Which of the following statements are true?

A. Unlike Ga that has a very high melting point, Cs has a very low melting point.

B. On Pauling scale, the electronegativity values of N and Cl are not the same.

C. Ar , K^+ , Cl^- , Ca^{2+} , and S^{2-} are all isoelectronic species.D. The correct order of the first ionization enthalpies of Na, Mg, Al, and Si is $\text{Si} > \text{Al} > \text{Mg} > \text{Na}$.

E. The atomic radius of Cs is greater than that of Li and Rb.

Choose the **correct** answer from the options given below :

(1) A, C, and E only

(2) A, B, and E only

(3) C and E only

(4) C and D only

Answer (3)

Sol. Both Ga and Cs have low melting points.

Element	Melting point/K
Ga	303
Cs	302

- On Pauling scale, the electronegativity value of N and Cl have same (3.0).
- Ar, K^+ , Cl^- , Ca^{2+} and S^{2-} have 18 electrons. So these are isoelectronic species.
- The correct order of first ionization enthalpy is $Si > Mg > Al > Na$
First ionisation enthalpy of Mg is higher than Al because the penetration of a 3s-electron to the nucleus is more than that of a 2p-electron.
- Generally down the group atomic radii increases

Atom	Atomic radius/pm
Li	152
Rb	244
CS	262

70. Given below are two statements :

Statement I : Like nitrogen that can form ammonia, arsenic can form arsine.

Statement II : Antimony cannot form antimony pentoxide.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- Statement I is incorrect but Statement II is correct
- Both Statement I and Statement II are correct
- Both Statement I and Statement II are incorrect
- Statement I is correct but Statement II is incorrect

Answer (4)

Sol. All the elements of group 15 form hydrides of EH_3 type. Nitrogen forms ammonia (NH_3) while Arsenic forms Arsine (AsH_3)

All the elements of group 15 form two types of oxides : E_2O_3 and E_2O_5

Antimony forms antimony pentoxide Sb_2O_5

Hence, statement I is correct and statement II is incorrect

71. Which of the following aqueous solution will exhibit highest boiling point?

- | | |
|---------------------------|----------------------|
| (1) 0.015M $C_6H_{12}O_6$ | (2) 0.01M Urea |
| (3) 0.01M KNO_3 | (4) 0.01M Na_2SO_4 |

Answer (4)

Sol.

$$\Delta T_b = iK_b \times m$$

$$\Delta T_b \propto i \times m$$

By considering molarity same as molality

$$(1) \quad 0.015 \text{ M C}_6\text{H}_{12}\text{O}_6 \quad i \times m = 1 \times 0.015 = 0.015$$

$$(2) \quad 0.01 \text{ M Urea} \quad i \times m = 1 \times 0.01 = 0.01$$

$$(3) \quad 0.01 \text{ M KNO}_3 \quad i \times m = 2 \times 0.01 = 0.02$$

$$(4) \quad 0.01 \text{ M Na}_2\text{SO}_4 \quad i \times m = 3 \times 0.01 = 0.03$$

$$T'_b = T_b^* + \Delta T_b$$

Higher the value of ($i \times m$) more will be the boiling point.

72. Given below are two statements :

Statement-I : Benzenediazonium salt is prepared by the reaction of aniline with nitrous acid at 273 – 278 K. It decomposes easily in the dry state.

Statement-II : Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

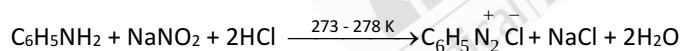
In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Statement I is incorrect but Statement II is correct
- (2) Both Statement I and Statement II are correct
- (3) Both Statement I and Statement II are incorrect
- (4) Statement I is correct but Statement II is incorrect

Answer (2)

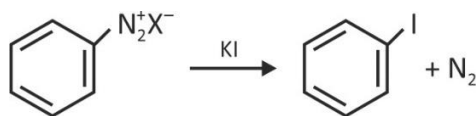
Sol. Benzene diazonium chloride is prepared by the reaction of aniline with nitrous acid at 273-278 K.

Nitrous acid is produced in the reaction mixture by reaction of NaNO_2 with HCl.

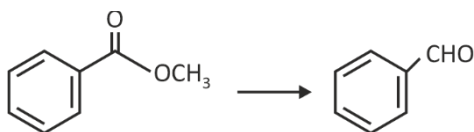


Benzene diazonium chloride decomposes easily in the dry state

Iodobenzene is prepared by shaking benzene diazonium salt with KI because direct insertion of iodine into benzene ring is difficult



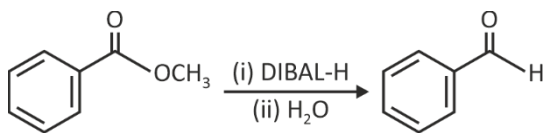
73. Identify the suitable reagent for the following conversion.



- | | |
|--|---|
| (1) $\text{H}_2/\text{Pd-BaSO}_4$ | (2) (i) LiAlH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$ |
| (3) (i) $\text{AlH}(\text{iBu})_2$, (ii) H_2O | (4) (i) NaBH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$ |

Answer (3)

Sol. Esters are reduced to aldehydes with DIBAL-H



74. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : undergoes $\text{S}_{\text{N}}2$ reaction faster than Cl.

Reason (R) : Iodine is a better leaving group because of its large size.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) **A** is false but **R** is true
- (2) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (3) Both **A** and **R** are true but **R** is **not** the correct explanation of **A**
- (4) **A** is true but **R** is false

Answer (2)

Sol. Rate of $\text{S}_{\text{N}}2$ reaction of I is faster than Cl.

Because iodine is a good leaving group due to large size of iodine. Which stabilises the I^- ion.

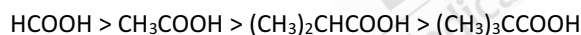
75. The correct order of decreasing acidity of the following aliphatic acids is

- (1) $\text{HCOOH} > (\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH}$
- (2) $(\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
- (3) $\text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH} > \text{HCOOH}$
- (4) $\text{HCOOH} > \text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$

Answer (4)

Sol. Electron donating group decreases the acidity of carboxylic acids.

So correct order is

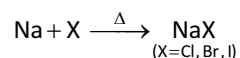
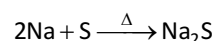
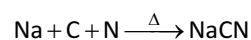


76. Which one of the following reactions does **NOT** belong to "Lassaigne's test"?

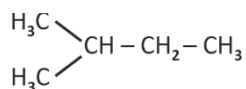
- (1) $2\text{CuO} + \text{C} \xrightarrow{\Delta} 2\text{Cu} + \text{CO}_2$
- (2) $\text{Na} + \text{C} + \text{N} \xrightarrow{\Delta} \text{NaCN}$
- (3) $2\text{Na} + \text{S} \xrightarrow{\Delta} \text{Na}_2\text{S}$
- (4) $\text{Na} + \text{X} \xrightarrow{\Delta} \text{NaX}$

Answer (1)

Sol. Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by "Lassaigne's test".



77. How many products (including stereoisomers) are expected from monochlorination of the following compound?

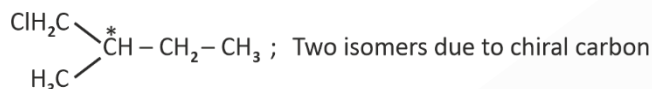
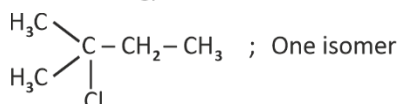
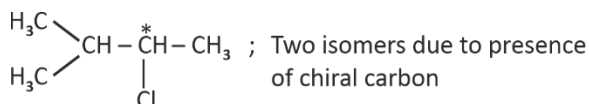
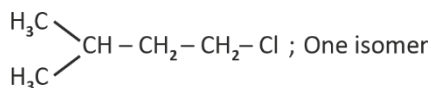


(1) 6 (2) 2

(3) 3 (4) 5

Answer (1)

Sol. Possible monochlorination products :



Total 6 isomers

78. Sugar 'X'

- A. is found in honey
B. is a keto sugar
C. exists in α and β – anomeric forms.
D. Is laevorotatory.

'X' is :

(1) Sucrose

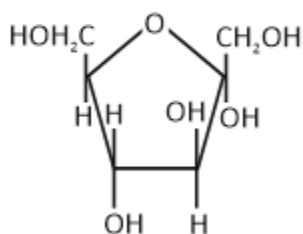
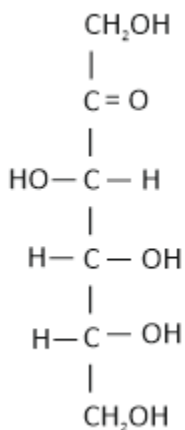
(2) D-Glucose

(3) D-Fructose

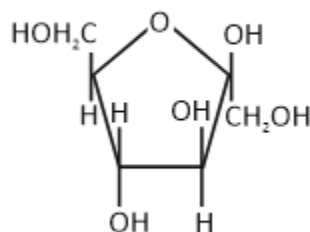
(4) Maltose

Answer (3)

Sol. D-Fructose is found in honey and is a keto sugar.



α -D-(-)-Fructofuranose



β -D-(-)-Fructofuranose

D-(-) — Fructose

79. Dalton's Atomic theory could not explain which of the following?

- | | |
|--------------------------------|---------------------------------|
| (1) Law of gaseous volume | (2) Law of conservation of mass |
| (3) Law of constant proportion | (4) Law of multiple proportion |

Answer (1)

Sol. Dalton's theory could explain the laws of chemical combination. However, it could not explain the laws of gaseous volumes.

80. Higher yield of NO in $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ can be obtained at

$[\Delta H \text{ of the reaction} = +180.7 \text{ kJ mol}^{-1}]$

- A. Higher temperature
B. Lower temperature
C. Higher concentration of N_2
D. Higher concentration of O_2

Choose the **correct** answer from the options given below :

- | | |
|------------------|------------------|
| (1) A, C, D only | (2) A, D only |
| (3) B, C only | (4) B, C, D only |

Answer (1)

Sol. Yield of the product generally depends on

- Temperature
- Concentration of reactant(s) and product(s)
- Pressure

As this is an endothermic reaction ($\Delta H = +180.7 \text{ kJ mol}^{-1}$), so, increase in temperature will shift equilibrium in forward direction to increase yield of NO.

Increase in concentration of reactants (N_2 and O_2) also shifts the equilibrium in forward direction and increase the yield of NO.

Hence, (A), (C) and (D) only will increase yield of NO.

81. Match **List-I** with **List-II**

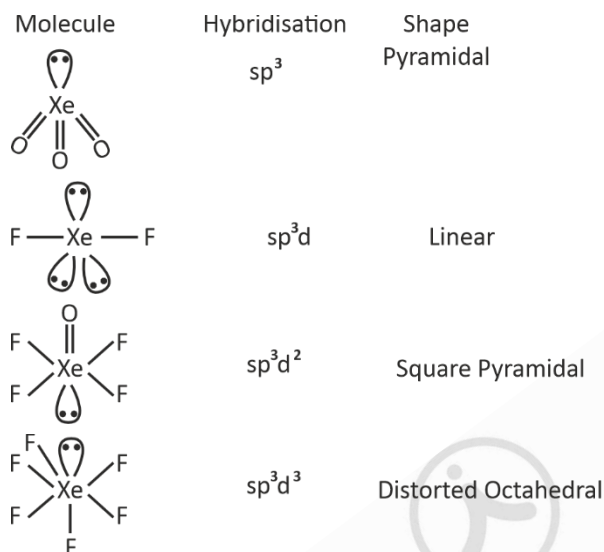
	List-I		List-II
A.	XeO_3	(I)	sp^3d ; linear
B.	XeF_2	(II)	sp^3 ; pyramidal
C.	XeOF_4	(III)	sp^3d^3 ; distorted octahedral
D.	XeF_6	(IV)	sp^3d^2 ; square pyramidal

Choose the **correct** answer from the options given below :

- (1) A-IV, B-II, C-I, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-IV, B-II, C-III, D-I

Answer (2)

Sol.



82. Match List-I with List-II

List-I

(Example)

- A. Humidity
- B. Alloys
- C. Amalgams
- D. Smoke

List-II

(Type of Solution)

- I. Solid in solid
- II. Liquid in gas
- III. Solid in gas
- IV. Liquid in solid

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-I, C-IV, D-II

Answer (3)

Sol.

- Humidity is a solution of liquid in gas
- Alloy is a solution of solid in solid
- Amalgam is a solution of liquid in solid
- Smoke is a solution of solid in gas

83. Energy and radius of first Bohr orbit of He^+ and Li^{2+} are

[Given $R_H = 2.18 \times 10^{-18} \text{ J}$, $a_0 = 52.9 \text{ pm}$]

(1) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-16} \text{ J}$;

$r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$

$E_n(\text{He}^+) = -19.62 \times 10^{-16} \text{ J}$;

$r_n(\text{He}^+) = 17.6 \text{ pm}$

(2) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-18} \text{ J}$;

$r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$

$E_n(\text{He}^+) = -8.72 \times 10^{-18} \text{ J}$;

$r_n(\text{He}^+) = 26.4 \text{ pm}$

(3) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-18} \text{ J}$;

$r_n(\text{Li}^{2+}) = 26.4 \text{ pm}$

$E_n(\text{He}^+) = -19.62 \times 10^{-18} \text{ J}$;

$r_n(\text{He}^+) = 17.6 \text{ pm}$

(4) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-16} \text{ J}$;

$r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$

$E_n(\text{He}^+) = -8.72 \times 10^{-16} \text{ J}$;

$r_n(\text{He}^+) = 26.4 \text{ pm}$

Answer (2)

Sol. $E_n = \frac{-2.18 \times 10^{-18} \times z^2}{n^2} \text{ J}$; $r_n = \frac{52.9 \times n^2}{z} \text{ pm}$

For He^+

$E_{\text{He}^+} = -2.18 \times 10^{-18} \times 4 = -8.72 \times 10^{-18} \text{ J}$

$r_{\text{He}^+} = \frac{52.9 \times 1}{2} = 26.45 \text{ pm}$

For Li^{2+}

$E_{\text{Li}^{2+}} = -2.18 \times 10^{-18} \times 9 = -19.62 \times 10^{-18} \text{ J}$

$r_{\text{Li}^{2+}} = \frac{52.9 \times 1}{3} = 17.63 \text{ pm}$

84. Which among the following electronic configurations belong to main group elements?

A. $[\text{Ne}]3s^1$

B. $[\text{Ar}]3d^34s^2$

C. $[\text{Kr}]4d^{10}5s^25p^5$

D. $[\text{Ar}]3d^{10}4s^1$

E. $[\text{Rn}]5f^06d^27s^2$

Choose the correct answer from the option given below :

(1) A, C and D only

(2) B and E only

(3) A and C only

(4) D and E only

Answer (3)

Sol. (A) $[\text{Ne}]3s^1$; Na (s-block)

(B) $[\text{Ar}]3d^34s^2$; V (d-block)

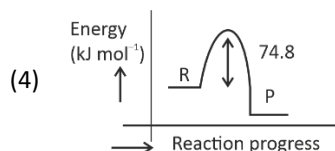
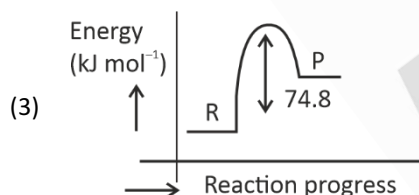
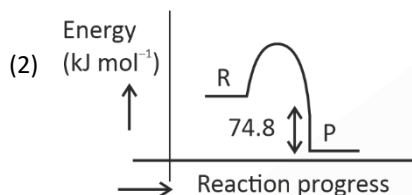
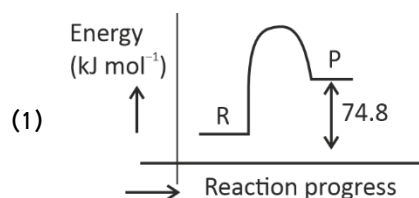
(C) $[\text{Kr}]4d^{10}5s^25p^5$; I (p-block)

(D) $[\text{Ar}]3d^{10}4s^1$; Cu (d-block)

(E) $[\text{Rn}]5f^06d^27s^2$; Th (f-block)

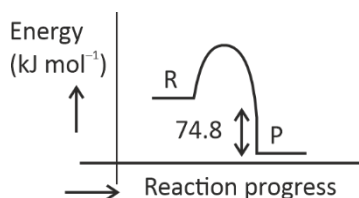
Main group elements (A and C only)

85. $\text{C(s)} + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g}); \Delta H = -74.8 \text{ kJ mol}^{-1}$. Which of the following diagrams gives an accurate representation of the above reaction? [R $\rightarrow$ reactants; P $\rightarrow$ products]

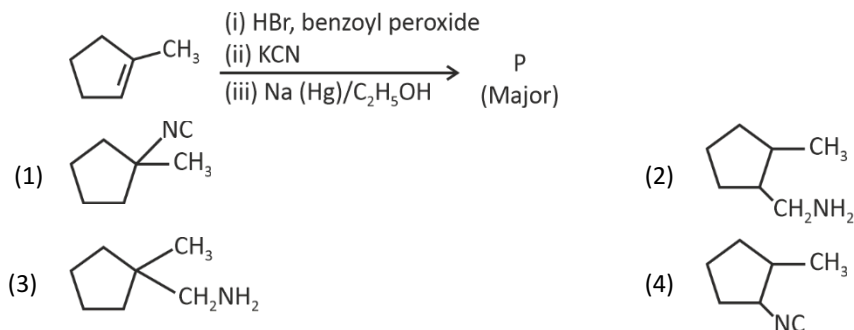
**Answer (2)**

Sol. $\Delta H = -74.8 \text{ kJ mol}^{-1}$, it is an exothermic reaction.

So, accurate representation is

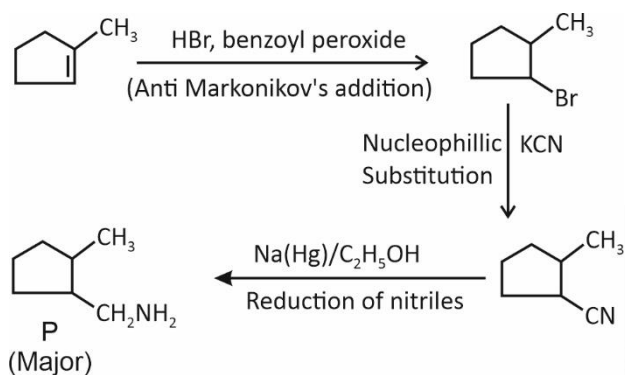


86. Predict the major product 'P' in the following sequence of reactions-



Answer (2)

Sol.



87. Identify the correct orders against the property mentioned

- $\text{H}_2\text{O} > \text{NH}_3 > \text{CHCl}_3$ – dipole moment
- $\text{XeF}_4 > \text{XeO}_3 > \text{XeF}_2$ – number of lone pairs on central atom
- $\text{O-H} > \text{C-H} > \text{N-O}$ – bond length
- $\text{N}_2 > \text{O}_2 > \text{H}_2$ – bond enthalpy

Choose the **correct** answer from the options given below:

- B, C only
- A, D only
- B, D only
- A, C only

Answer (2)

Sol.

	$\mu(\text{D})$
A. H_2O	1.85
NH_3	1.47
CHCl_3	1.04
B. XeF_4 : 2 lone pairs of electron	
XeO_3 : 1 lone pair of electron	
XeF_2 : 3 lone pairs of electron	

C. Order of Bond length :- $\text{N-O} > \text{C-H} > \text{O-H}$

D. N_2 Bond order is 3

H_2 Bond order is 1

O_2 Bond order is 2

88. Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula $\text{C}_4\text{H}_8\text{O}$ is :

(1) 11

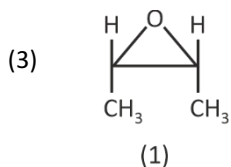
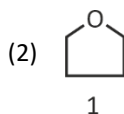
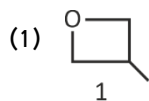
(2) 6

(3) 8

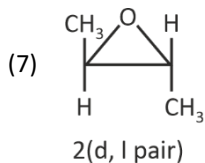
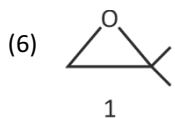
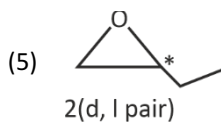
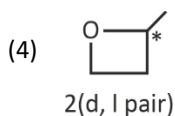
(4) 10

Answer (4)

Sol. For cyclic ethers O should be in ring * carbon here is chiral



Meso compound



Total number of isomers = $2 + 1 + 1 + 1 + 2 + 1 + 2 = 10$